The Valid Dictionary: Newtonian Mechanics

Can Valid

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Valid Dictionary (VD) Document: Newtonian Mechanics as a Valid Dictionary	**Status:** DRAFT
Internal ID: `vd-newton`	**Project tag:** `vd-project`

Point. Present a complete, minimal Newtonian mechanics encoding as a VD instance, demonstrating auditability and explicit structure.

Target goals.

- Provide a finished Newtonian VD with clear entry indexing and cross-references.
- Encode the relevant Newtonian "laws" as VD triplets consistent with `vd-core`.
- Keep commentary minimal: enough to read and verify the dictionary, not to persuade.

Non-goals.

- Arguing why the case study is important (belongs in `vd-newton-analysis`).
- Introducing new VD rules or changing the core spec.
- Covering advanced Newtonian topics beyond "basics" needed for demonstration.

Target audience. Readers who want to see VD used on real physics; skeptics who want something concrete to inspect.

Dependencies / references. Depends on `vd-core` for definitions, notation, and validity criteria.

1 Introduction

2 Current Dictionary Log (Finalised Entries So Far)

2.1 Non-law kinematics scaffold

E1 `time` := "A real parameter $t \in \mathbb{R}$ used to order events; differences Δt are durations measured by a clock."

E2 `reference_frame` := "A convention for assigning spatial coordinates and a time coordinate t to events: an origin, axes, and a clock."

E3 `position` := "In a reference_frame, an object's location at time t represented by a vector $\mathbf{r}(t) \in \mathbb{R}^3$."

E4 `displacement` := "$\Delta\mathbf{r} = \mathbf{r}(t_2) - \mathbf{r}(t_1)$."

E5 `trajectory` := "The time-ordered curve $\{\mathbf{r}(t) : t \in I\}$ for some interval $I \subset \mathbb{R}$."

E6 `velocity` := "$\mathbf{v}(t) = d\mathbf{r}/dt$."

E7 `speed` := "$v(t) = \|\mathbf{v}(t)\|$."

E8 `acceleration` := "$\mathbf{a}(t) = d\mathbf{v}/dt = d^2\mathbf{r}/dt^2$."

E9 `path_length` := "From t_1 to t_2: $s = \int_{t_1}^{t_2} \|\mathbf{v}(t)\|\, dt$."

E10 `point_particle` := "An idealized object whose size and internal structure are neglected; its motion in a reference_frame is fully represented by a trajectory $\mathbf{r}(t)$."

E11 `straight_line` := "A set $\mathbf{r}(\lambda) = \mathbf{r}_0 + \lambda\mathbf{u}$ with fixed $\mathbf{r}_0$, nonzero direction $\mathbf{u}$, and $\lambda \in \mathbb{R}$."

2.2 Uniform motion intermediates (non-law)

E12 `uniform_motion_on_interval(I)` := "A point_particle is in uniform_motion_on_interval(I) if its velocity is the same at every time in I; i.e. $\forall t_1, t_2 \in I,\ \mathbf{v}(t_1) = \mathbf{v}(t_2)$."

E13 `uniform_motion_at_time(t_0)` := "A point_particle is in uniform_motion_at_time(t_0) if $\exists \varepsilon > 0$ such that $\forall t \in (t_0 - \varepsilon, t_0 + \varepsilon)$, $\mathbf{v}(t) = \mathbf{v}(t_0)$."

2.3 Newton I (triplet: uniform motion as bridge; no acceleration)

E14 `uniform_motion_on_interval(I)` := "The state of motion exhibited by a free_particle when described in an inertial_frame over the interval I."

E15 `inertial_frame` := "A reference_frame in which every free_particle exhibits uniform_motion_at_time(t) (equivalently: uniform_motion_on_interval(I) on any interval of free motion)."

E16 `free_particle` := "A point_particle that, when described in an inertial_frame, exhibits uniform_motion_at_time(t) (equivalently: uniform_motion_on_interval(I)) whenever treated as free."

2.4 Newton II Law

E17 `acceleration` := "For a point_particle described in an inertial_frame, acceleration equals net_force divided by inertial_mass: $\mathbf{a} = \mathbf{F}_{\texttt{net}}/\texttt{inertial_mass}$ (equivalently: $\mathbf{F}_{\texttt{net}} = \texttt{inertial_mass} \cdot \mathbf{a}$)."

E18 `net_force` := "The vector quantity satisfying `net_force` = `inertial_mass` · `acceleration` for the point_particle (in an inertial_frame)."

E19 `inertial_mass` := "The positive scalar coefficient m such that `net_force` = $m \cdot$ `acceleration` for the point_particle (in an inertial_frame)."

2.5 Force sum Law (triplet: acting forces and interaction set)

E20 `net_force(p,t)` := "For any particle p and time t, net_force(p, t) is the vector-sum of all acting_force elements in interacting_forces_set(p, t) (equivalently: $\texttt{net_force}(p,t) = \sum_{F \in \texttt{interacting_forces_set}(p,t)} F$)."

E21 `interacting_forces_set(p,t)` := "The set of acting_force vectors that the modeler judges to be acting on particle p at time t (i.e. the force-contributions currently included for p at t)."

E22 `acting_force` := "A vector force-contribution F that appears as an element of interacting_forces_set(p, t) for some particle p and time t."

2.6 Meta-typing (triplet: raw classes and recognition-based instantiation)

E23 `raw_class` := "An abstract object-type."

E24 `raw_class` := "An abstract object-type specified by class_specific_features(C); an object x is treated as instance_of(x, C) when it is recognized as possessing the class_specific_features(C)."

E25 `class_specific_features(C)` := "The set/list of features that specify raw_class C; these features are whatever a modeler must recognize x as having in order to treat x as instance_of(x, C)."

E26 `instance_of(x,C)` := "x is treated as an instance of raw_class C when x is recognized as possessing class_specific_features(C)."

2.7 Action-Reaction Law(not Newton III)

E26 `Acting force.` := "*Acting force* is a force–contribution that is either a *canonical_force(acting_object)* or a *reaction_force(acting_object)*."

E28 `Canonical force.` := "A *canonical_force(acting_object)* is an acting force designated as canonical; the corresponding *reaction_force(acting_object)* is the acting force defined as the negative of it."

E29 `Reaction force.` := "A *reaction_force(acting_object)* is an acting force defined as the negative of *canonical_force(acting_object)*."

2.8 Acting object Law

E30 `acting_object` := An *acting_object* is an object equipped with both an *activation_condition(acting_object)* and a corresponding *canonical_force(acting_object)*. (This is a class)

E31 `Activation condition.` := If a particle fulfills the *activation_condition(acting_object)* the *canonical_force(acting_object)* will be an acting-force on the particle.

E32 `Canonical force (of an acting object).` := The *canonical_force(acting_object)* is the acting-force of a particle that fulfills the *activation_condition(acting_object)*.

2.9 Non-law

E33 `paired_particle(acting_object)` := Each *acting_object* carries an associated particle, called its *paired_particle*. The acting object is considered to be located at the location of its paired particle .The particle may act as the object's COM.

E34 `mechanical_system` := An arbitrary subset of particles $\{p_1, \ldots, p_n\}$ selected by the observer for analysis.

E35 `interaction_candidate` := For a particle p in a mechanical system S at time t, an *interaction candidate* is any acting force in $interacting_forces_set(p, t)$.

2.10 Mechanical Closure Law

E36 `interaction_candidate` := An *interaction candidate* is an elementary force-unit, every one of which must be able to be placed in an *interaction pair* for the mechanical system to be a *mechanically closed system*.

E37 `interaction_pair` := An *interaction pair* is the form into which *interaction candidates* are placed in a *mechanically closed system*.

E38 `mechanically_closed_system` := A *mechanically closed system* is a *mechanical system* in which every *interaction candidate* can be placed in an *interaction pair*.

2.11 Non-law

E39 `interaction_pair` := Two *interaction candidates* form an *interaction pair* when they correspond to the same *acting_object*, one being the *canonical_force* associated with that acting object and the other being the *reaction_force* associated with that acting object, acting on the object's *paired_particle*.

E40 `mechanically_closed_system` := A *mechanically closed system* is a *mechanical system* whose future behaviour is determined by the equations produced from the interactions internal to it.